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RESEARCH ARTICLE

Investigating the Impact of Rational Dilated Wavelet Transform on Motor Imagery EEG Decoding With Deep Learning Models

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ABSTRACT Decoding neural signals from electroencephalogram (EEG) data is a challenging task due to the signals’ intrinsic non-stationarity, low signal-to-noise ratio, and complex spatio-temporal dynamics. The present study investigates the impact of the Rational Dilated Wavelet Transform (RDWT) for implementing a de-noising operation before applying deep learning classifiers devised to recognize Motor Imagery (MI) EEG signals. A systematic paired evaluation (with/without RDWT) is conducted on four state-of-the-art deep learning architectures and carried out across three benchmark datasets. The performance of the models integrating RDWT is reported with subject-wise averages using accuracy and Cohen’s kappa, complemented by subject-level analyses to identify when RDWT is beneficial. Our results show improvements in classifier performance in almost all cases, and especially for MI tasks/trials that are difficult to classify. We then conclude that RDWT-based signal preprocessing can mitigate localized noise and enhance the EEG classification performance without adding significant complexity to the classifier architecture.

INDEX TERMS EEG decoding, rational dilated wavelet transform, brain-computer interfaces, neural signal processing, deep learning, time-frequency analysis.

I. INTRODUCTION

Decoding electroencephalographic (EEG) signals for Brain-Computer Interfaces (BCIs) remains challenging due to low signal-to-noise ratio [1], [2] and pronounced inter-/intra-subject variability [3], [4]. Variations in emotional and physiological states further inject artifacts, making robust noise suppression and adaptive time-frequency feature extraction essential [5], [6].

Several approaches to BCI decoding have been explored so far, either in terms of signal representation in time, frequency, time-frequency domains, or in terms of classifier model and architectures. In [7] it has been shown that for deep neural

networks it is beneficial to use a signal representation in the time domain. In other words, much of the contemporary Motor Imagery (MI) EEG decoding literature avoids the signal preprocessing stage. The prevailing approach is to learn *end-to-end* from raw or simply resampled signals, without explicit denoising for artifact suppression [7]. In practice, this leaves ocular and muscular artifacts, power-line interference, baseline drift, and marked cross-subject/cross-session variabilities in the input space. These contaminants can mask idiosyncratic, subject-dependent spatio-spectral signatures (e.g., shifts in μ/β peaks and individualized temporal dynamics) that drive MI performance, forcing the model to allocate capacity to nuisance variance rather than class-discriminative structure and ultimately reducing robustness and out-of-subject generalization at test time.

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In this paper, we explore the possibility of adding a pre-processing operation based on *Rational Dilated Wavelet Transforms* (RDWTs), before performing the signal classification. Indeed, RDWT allows processing different signal sub-bands, improving de-noising and rhythm-specific enhancement in MI settings. Rather than proposing a new classifier, our goal is to *quantify if, when and how much* the usage of RDWT for filtering improves the performance across heterogeneous models and datasets. Differently from recent studies that explore the potential of RDWT filtering for MI signal classification using classical machine learning models [8], this work investigates the impact on modern deep learning architectures. By systematically evaluating the impact of RDWT on widely-used deep learning models, we provide crucial insights into its utility as an architecture-agnostic preprocessing step for advanced EEG decoding. Our study addresses the question of whether RDWT should become a standard component in the pipeline for deep learning-based MI classification.

We conduct a systematic, paired comparison (with and without RDWT) on four state-of-the-art deep learning architectures, namely EEGNet [9], ShallowConvNet [10], MBEENet [11] and EEGTCNet [12] on three benchmark datasets: High Gamma (HGD) [13], BCI Competition IV-2a [14], and IV-2b [15]. Performance is assessed with subject-wise averages using accuracy and Cohen's κ . Beyond aggregate metrics, we analyse subject-level heterogeneity to identify conditions under which RDWT is particularly beneficial (e.g., challenging MI samples).

A. PREVIEW OF FINDINGS

RDWT's effect is *architecture- and dataset-dependent*. On BCI-IV-2a, RDWT tends to improve performance of models with explicit temporal stacks (e.g., TCN-based backbones), while effects on already multi-scale CNN pipelines are negligible to modest. On BCI-IV-2b, several subjects approach ceiling accuracy, yielding small average changes despite occasional subject-level gains. On HGD, average effects are modest but RDWT can materially help specific difficult subjects, suggesting its value as a targeted de-noising/preprocessing step. The use of RDWT is a low-cost preprocessing step that yields practical gains in the classification performance of both traditional machine learning [8] and deep learning models.

This study is guided by the following research questions (RQs):

- **RQ1.** Given the prevailing trend of end-to-end learning from raw EEG signals, what is the practical value and performance gain yielded by introducing the RDWT as a dedicated de-noising and feature-enhancement step prior to deep learning classification?
- **RQ2.** Is the benefit of the RDWT uniform across all deep learning architectures, or is it selectively maximized for models specialized in capturing temporal dynamics (e.g., TCNs)?

In summary, the paper reframes the process of EEG pre-processing as a design choice of paramount importance - often neglected in recent deep learning literature - and offers a study for the impact of RDWT in modern pipelines. Our contributions are:

- A cross-architecture, cross-dataset evaluation of RDWT as a plug-in preprocessing step for EEG decoding before employing advanced deep learning architectures, with paired (with/without) comparisons on HGD, BCI-IV-2a, and BCI-IV-2b.
- A subject-wise analysis that reveals when RDWT is most beneficial (e.g., challenging subjects, non-stationary conditions, and temporal-modeling backbones), alongside aggregate accuracy and Cohen's κ .
- Practical guidance for integrating RDWT in BCI pipelines, highlighting scenarios with expected payoff versus near-ceiling regimes where benefits are limited.

II. RELATED WORK

A. TIME-FREQUENCY ANALYSIS IN EEG

The EEG signal, which reflects cortical electrical activity, is inherently susceptible to various sources of noise and artifacts, including cardiac activity, muscular contractions (electromyography, EMG), and ocular movements (electrooculography, EOG). Despite these contaminants, motor intention is characterized by distinct spectral phenomena within the sensorimotor cortex. Specifically, the imagination or execution of movement modulates the μ (7–13 Hz) and β (13–30 Hz) rhythms. The onset of motor activity is associated with a decrease in the amplitude of these oscillations, a phenomenon known as Event-Related Desynchronization (ERD). Conversely, the cessation or resting state is typically marked by an increase in amplitude, particularly in the β band, termed Event-Related Synchronization (ERS). These transient ERD/ERS dynamics form the essential discriminative features utilized for MI decoding [16].

Traditional signal processing techniques have long served as the foundation of EEG-based BCIs, offering a means to extract time/frequency-domain features from inherently non-stationary neural signals. The STFT is widely used to obtain time-localized spectral information via a sliding-window Fourier transform. However, its reliance on a fixed window size inherently limits the trade-off between time and frequency resolution [17], making it less suited for tracking rapid transients in EEG.

Wavelet-based approaches, particularly the DWT, have demonstrated improved performance for EEG decoding [18] by enabling multi-resolution analysis of signal components. DWT is particularly effective in isolating oscillatory activity across characteristic EEG frequency bands [19], and has been used both as a standalone preprocessing step and in conjunction with machine learning classifiers.

Recent advances have combined DWT with deep learning architectures, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), where wavelet-decomposed EEG signals serve as inputs to enhance

discriminative feature extraction [20]. Despite these improvements, both STFT and conventional DWT employ fixed or integer-valued dilation/scaling factors, which can fail to capture the variable temporal dynamics and scale-dependent structures present in EEG recordings.

While recent research has showcased the efficacy of RDWT in enhancing MI signal classification, these efforts have predominantly relied on conventional machine learning models employed for the classification [8]. The present study takes a distinct approach by bridging the gap between the RDWT-based preprocessing and modern deep learning architectures. Instead of simply extracting features to be provided as input to traditional classifiers, our work systematically evaluates how RDWT integration impacts the performance of state-of-the-art deep models designed for EEG decoding. We aim to provide a comprehensive analysis of whether this approach can yield tangible improvements, thereby establishing a strong case for including RDWT as a standard preprocessing step in deep learning-based BCI pipelines.

B. RDWT PREPROCESSING

The RDWT [21], via *rational* dilations p/q , offers a more flexible multiresolution decomposition than dyadic DWT, enabling finer alignment of sub-bands to μ/β oscillations and improved capture of short bursts while preserving temporal structure required by temporal backbones (e.g., TCN/CNN). RDWT uses non-integer dilation factors, such as $3/2$ or $5/3$, to adapt the frequency resolution to the underlying signal structure more effectively. This provides a better time-frequency localization for non-stationary signals, making it particularly suitable for EEG preprocessing in BCI applications. Applying a *soft-threshold* to detail coefficients implements principled wavelet shrinkage, attenuating EMG/EOG contamination and high-frequency noise while limiting undue smoothing of informative onsets/offsets. Compared with STFT, RDWT avoids fixed-window trade-offs and reduces leakage on brief transients; compared with data-driven decompositions (e.g., EMD), it is deterministic, lightweight, and easy to integrate into reproducible training/inference pipelines. In this study we use RDWT as a *model-agnostic* front-end with a single configuration held fixed across models and datasets to isolate its effect; to contextualize the results, we additionally include a four level Daubechies Wavelet (DB4) baselines with *soft* and *hard* coefficient thresholding (DB4-Soft/DB4-Hard) under identical splits and hyperparameters, and we report paired (Wilcoxon) p-values for all comparisons against RDWT.

The input of the RDWT is a segment of the EEG signal, formally defined as $x_{e,c}(t)$. Here, the subscript e denotes a specific experimental **event** or trial (e.g., a single instance of imagined left-hand movement), which belongs to the set of all trials $E = \{1, \dots, N_E\}$. The subscript c represents the electrode **channel** or spatial location on the scalp, belonging to the set $C = \{1, \dots, N_C\}$, corresponding to the total number of recorded channels. Finally, t is the discrete

time index, where $t \in \{1, \dots, T\}$, represents the sampled time points within the duration of the trial segment. The RDWT decomposes $x_{e,c}(t)$ into a set of sub-bands across multiple resolution levels $l = 1, \dots, L$, using rational dilation factors $d_l \in \mathbb{Q}$. The decomposition is performed as follows:

$$d_{e,c}^{(l)}(t) = \left(x_{e,c}^{(l-1)} * h_{\text{low}}^{(l)} \right)(t), \quad d_{e,c}^{(l)}(t) = \left(x_{e,c}^{(l-1)} * h_{\text{high}}^{(l)} \right)(t), \quad (1)$$

where $x_{e,c}^{(0)}(t) = x_{e,c}(t)$ and $*$ denotes the linear convolution. The filters $h_{\text{low}}^{(l)}$ and $h_{\text{high}}^{(l)}$ are rationally resampled versions of standard wavelet filters:

$$h^{(l)}(n) = \text{Resample}(h(n), \text{scale} = d_l), \quad (2)$$

where $d_l \in \left\{ \frac{3}{2}, \frac{5}{3}, \frac{7}{4}, \frac{9}{5} \right\}$. This multiresolution approach ensures translation invariance, which is critical for time-localized EEG features.

To suppress high-frequency noise, the detail coefficients are thresholded:

$$\tilde{d}_{e,c}^{(l)}(t) = \begin{cases} d_{e,c}^{(l)}(t), & \text{if } |d_{e,c}^{(l)}(t)| \geq \tau, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

where τ is a fixed amplitude threshold. The final signal is reconstructed by summing the final approximation and all thresholded detail components:

$$x_{e,c}^{(l)}(t) \leftarrow a_{e,c}^{(l)}(t), \quad \hat{x}_{e,c}(t) = a_{e,c}^{(L)}(t) + \sum_{l=1}^L \tilde{d}_{e,c}^{(l)}(t). \quad (4)$$

This preprocessing module ensures a denoised, translation-invariant multiresolution representation of the original EEG input, suitable for downstream deep learning models.

1) RDWT: PARAMETERIZATION AND BRIEF SENSITIVITY

We employ a four-level RDWT front-end. Rational dilations are initialized from a single base scale s_0 , yielding level-wise dilations $r_\ell = s_0 \cdot \alpha_\ell$ with a fixed, increasing set $\{\alpha_\ell\}_{\ell=1}^4$; unless otherwise noted, these dilations are kept fixed (no search). The soft-threshold τ controls coefficient shrinkage/gating and is selected on the validation split via a one-dimensional grid search.

III. THE RDWT MODULE: THEORETICAL FRAMEWORK AND MOTIVATION

The selection of the RDWT as our feature-enhancing front-end is grounded in the specific requirements of MI EEG signal processing.

A. THEORETICAL RATIONALE FOR RDWT

Unlike DWTs that use a fixed mother wavelet (e.g., Daubechies) and binary scaling, RDWT utilizes a rational filter bank that offers several key advantages for non-stationary signals like EEG:

- 1) Enhanced Frequency Resolution: The rational dilation factor provides finer control over the frequency tiling,

allowing for a more accurate separation of closely located frequency components (like the μ and β bands) compared to standard DWTs.

- 2) **Flexibility and Optimization:** The filter coefficients in RDWT are designed to achieve optimal time-frequency localization, which minimizes spectral leakage and enhances the quality of the resulting wavelet coefficients.

B. JUSTIFICATION OF PARAMETERS

The specific parameters for our RDWT implementation were chosen to maximize the extraction of the motor imagery-relevant frequency components:

- **Mother Wavelet/Filter Structure:** We adopt the rational filter bank structure because its design is inherently superior at minimizing inter-band energy leakage, making it more effective at isolating the target rhythms than conventional fixed wavelets (like Db4, as shown in our comparative analysis).
- **Decomposition Level ($L = 4$):** The decomposition level was specifically chosen to span the critical 8–30 Hz range relevant for MI decoding. A four-level decomposition is sufficient to span the required frequency scales, balancing the need for granular feature extraction against computational complexity. Lower decomposition levels would miss fine-grained spectral detail, while excessively high levels introduce redundant or irrelevant features.
- **Thresholding Strategy:** The soft thresholding approach is adopted because it guarantees a continuous signal reconstruction, which is critical for preserving the subtle temporal dependencies required by the subsequent deep learning backbones (especially TCN-based models).

By selecting RDWT and its parameters based on these theoretical considerations, we ensure that the input to the deep learning classifiers is maximally denoised and rich in task-relevant spectral features.

C. THEORETICAL COMPARISON WITH ALTERNATIVE TRANSFORMS

The choice of RDWT over other established time-frequency representations—specifically the discrete wavelet transform (DWT), continuous wavelet transform (CWT), and tunable Q-factor wavelet transform (TQWT)—is driven by a trade-off between spectral resolution, computational efficiency, and feature sparsity required by deep learning backbones.

1) RDWT VS. STANDARD DWT

The classical DWT employs a dyadic grid (powers of two), which rigidly partitions the frequency spectrum. This often leads to *spectral leakage* when the physiological bands of interest (e.g., μ at 8–13 Hz or β at 13–30 Hz) straddle the boundaries of dyadic sub-bands. RDWT mitigates this by allowing rational dilation factors (e.g., p/q), enabling a flexible tiling of the time-frequency plane that can be tuned

to align more closely with specific EEG rhythms without the constraints of octave-based scaling. Furthermore, unlike the critically sampled DWT, the RDWT formulation used here avoids shift-variance, ensuring that the extracted features remain robust to small temporal translations of the motor imagery onset.

2) RDWT VS. CWT

While the CWT offers high-resolution scalograms (as visualized in Fig. 1) and is widely used for visual inspection, it is highly redundant. CWT generates dense 2D images from 1D signals, significantly increasing the input dimensionality. For deep learning models, this explosion in dimensionality often leads to the “curse of dimensionality,” requiring larger models and increasing the risk of overfitting, as well as imposing a prohibitive computational burden for real-time BCI applications. RDWT provides a sparse, multi-level representation that captures the essential oscillatory dynamics without the extreme redundancy of CWT, preserving the low latency required for online decoding.

3) RDWT VS. TUNABLE Q-FACTOR WAVELET TRANSFORM (TQWT)

The TQWT allows for the adjustment of the Q-factor (the ratio of center frequency to bandwidth) to match the oscillatory behavior of the signal. While theoretically attractive for separating oscillatory components from transients, TQWT involves a complex parameterization (Q-factor, redundancy, and number of stages) that is difficult to optimize universally across varying subjects. Moreover, the synthesis and analysis banks of TQWT are computationally more demanding than the rational filter banks of RDWT. RDWT offers a pragmatic balance: it provides sufficient flexibility to capture transient motor imagery bursts through rational dilations, yet maintains a deterministic and lightweight structure that integrates seamlessly as a pre-processing layer in efficient convolutional neural networks.

In summary, RDWT is selected as the optimal “middle ground”: it surpasses DWT in spectral isolation capabilities and outperforms CWT and TQWT in terms of computational efficiency and representational compactness suitable for deep neural architectures.

IV. EXPERIMENTAL SETUP

All experiments were conducted on a workstation equipped with an Intel® Xeon W7-3455 processor, an NVIDIA RTX 6000 Ada Generation GPU featuring 48 GB of memory, and 187 GiB of DDR5 system RAM. Training durations varied according to the dataset size and complexity: approximately 70 minutes for the High-Gamma Dataset, and around 42 minutes for both BCI-IV-2a and BCI-IV-2b datasets. Such variation reflects the inherent differences in data volume and complexity among the datasets. For all experiments, we adopted a consistent training protocol: optimization was performed using the Adam optimizer with a learning rate of 0.001, batch size of 64, and training

proceeded for a maximum of 500 epochs, with early stopping triggered after 100 epochs without improvement. Our evaluation was performed on three publicly available EEG benchmark datasets: BCI-IV-2a [14], BCI-IV-2b [15], and the High Gamma Dataset [13]. These datasets differ in acquisition hardware, experimental protocols, and number of subjects, providing a robust assessment of the model's generalization capabilities. The code and experimental results are publicly available on GitHub.

A. DATASET DESCRIPTION

In this section, we present the characteristics of the datasets utilized in our experiments.

1) BCI IV 2A DATASET

The BCI Competition IV Dataset 2a [14] is a well-established resource for evaluating MI classification models. It includes EEG data collected from nine healthy volunteers (subjects A01–A09), each participating in two separate sessions recorded on different days to assess session variability. Participants were instructed to imagine the movement of one of four distinct body parts: left hand, right hand, both feet, or tongue. These actions were selected due to their distinct cortical representations, particularly in sensorimotor regions. Each session includes 288 randomized trials (72 per class). The experimental protocol followed a cue-based structure, starting with a fixation cross, followed by a directional arrow indicating the target MI class. EEG signals were recorded via 22 Ag/AgCl electrodes following the 10–20 system, at a sampling rate of 250 Hz, and band-pass filtered between 0.5–100 Hz. Additionally, three EOG channels were recorded for ocular artifact correction. Signals were referenced to the left mastoid and grounded at AFz. Recordings were carried out in an electromagnetically shielded environment to minimize interference. This dataset is maintained by the Graz BCI Laboratory and has become a reference benchmark in motor imagery decoding, often used to test methods like CSP, deep learning models, and geometric classifiers.

2) BCI IV 2B DATASET

The BCI Competition IV Dataset 2b [15] is another standard benchmark tailored for binary-class MI studies, specifically focused on distinguishing between left- and right-hand imagery. EEG data were collected from nine healthy individuals (B01–B09) across five sessions. Sessions 1 and 4 were conducted without feedback, while the remaining sessions included real-time feedback. Each session consisted of 160 trials (80 per class), summing to 800 trials per subject. During each trial, subjects performed kinesthetic imagery of a hand movement in response to an arrow cue. EEG was recorded from three bipolar channels (C3, Cz, and C4) targeting the sensorimotor cortex, using a 250 Hz sampling rate and a band-pass of 0.5–100 Hz, with a 50 Hz notch filter to remove powerline noise. A minimal channel setup was used to emulate portable BCI scenarios. Feedback, when available, was presented as a bar indicating

model predictions. Due to its simplicity and challenge, BCI-2B is frequently used to validate algorithms for motor imagery classification, including CSP variants and deep neural models.

3) HIGH-GAMMA DATASET

The High-Gamma Dataset (HGD) [13] contains high-density EEG recordings from 14 healthy participants (6 female, 2 left-handed, mean age 27.2 ± 3.6 years). Data were collected using 128 electrodes, with analysis focused on 44 channels covering the motor cortex. Subjects performed four tasks, imagined movements of the left hand, right hand, feet, and a resting state, over 13 runs, each comprising 80 trials of 4 seconds. On average, around 963 trials were recorded per subject, with the final two runs reserved for testing. Visual cues guided the tasks using directional arrows, with rest indicated by an upward arrow. The experimental setup emphasized minimal muscular movement, focusing on tasks like finger or toe tapping. Data acquisition was performed in a shielded environment with low-noise amplifiers and active shielding, and signals were recorded using the BCI2000 system at a sampling rate of 5 kHz [22]. The study was ethically approved by the University of Freiburg and is widely used to benchmark deep learning-based EEG decoding techniques.

B. EVALUATION METRICS

The performance of the preprocessing in the models is assessed using two standard metrics: accuracy and the Cohen's Kappa coefficient. These metrics are widely adopted in the EEG-based classification literatures [23], [24], and [25] for benchmarking purposes. Accuracy is defined as the proportion of correctly classified instances over the total number of samples. Cohen's Kappa quantifies the agreement between predicted and true labels while accounting for the agreement occurring by chance. The definitions of Accuracy and Kappa are given by Equation (5):

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n \frac{TP_i}{I_i}, \quad \text{Kappa} = \frac{1}{n} \sum_{a=1}^n \frac{P_a - P_e}{1 - P_e} \quad (5)$$

where TP_i denotes the number of correctly predicted samples for class i , I_i is the total number of samples in class i , n is the total number of classes, P_a is the observed accuracy across classes, and P_e is the expected accuracy under random chance.

C. DEEP BACKBONES AND TRAINING PROTOCOL

We evaluate four commonly used architectures and feed each of them with the same inputs under two paired conditions (baseline and RDWT). EEGNet is employed in its canonical form with depthwise-separable temporal convolution at the front end and subsequent spatial filtering across channels; batch normalization and dropout stabilize training, while the compact temporal receptive field makes the model particularly sensitive to the quality of the input envelopes.

ShallowConvNet follows the shallow, filter-bank inspired design with an initial temporal convolution that approximates band-pass extraction, a spatial filtering stage across sensors, and a sequence of squaring, mean pooling and logarithmic compression, which together favour smooth envelope information over high-frequency transients. MBEEG_SENet adopts a multi-branch convolutional stem that aggregates features at different temporal scales and reweights channel responses via squeeze-and-excitation, thus emphasising spatially coherent rhythms when they are reliably expressed. EEGTCNet builds on temporal convolutional blocks with residual connections and dilations; by virtue of its larger effective temporal receptive field, it is able to integrate the multiscale cues exposed by the RDWT representation more extensively than the compact CNNs.

All models are trained with the same protocol so that differences can be ascribed to the front end and the backbone rather than to idiosyncratic optimization. Unless otherwise stated, we use Adam with a fixed learning rate of 0.001, a batch size of 64, and a maximum of 500 epochs; model selection is performed on the best validation accuracy, and the final report includes subject-wise accuracy and Cohen's κ . Randomness is controlled by fixed seeds for NumPy and PyTorch with deterministic back ends enabled; the software stack and hardware remain constant across runs. The RDWT condition replaces the raw series with a four-level rationally dilated wavelet representation that tiles the μ/β range (8–30 Hz) with mild overlap, using a zero-mean, unit-energy, approximately analytic prototype and a soft-threshold selected on the validation split. The baseline condition retains the very same epochs and labels and applies only per-channel z-scoring computed on the training split. Exact hyperparameters for each dataset–model pair (including kernel sizes, number of filters per stage, dropout rates and dilation schedules where applicable) as well as per-subject notebooks with training/validation traces, confusion matrices and per-class bar plots are provided in the companion repository to permit end-to-end replication.

D. BASELINE COMPARISON

For comparative evaluation, we selected a set of state-of-the-art models commonly employed in EEG classification tasks: *EEGNet* [9], *ShallowConvNet* [10], *MBEEG_SENet* [11], and *EEG-TCNet* [12]. Each baseline was trained using the original hyperparameters reported in their respective works. A standardized preprocessing, training, and evaluation pipeline was employed across all models to ensure fairness in comparison.

Briefly, *EEGNet* utilizes a combination of 2D temporal, depthwise, and separable convolutions tailored for general BCI classification tasks. *ShallowConvNet* comprises two convolutional layers followed by mean pooling operations optimized for EEG decoding. *MBEEG_SENet* incorporates a multi-branch convolutional architecture with squeeze-and-excitation blocks to capture multi-scale EEG features.

EEG-TCNet combines CNNs and TCNs to effectively model temporal dependencies.

V. RESULTS

This section presents a comprehensive, paired evaluation of the RDWT as a plug-in preprocessing step across multiple EEG classifiers. For each model (*EEGNet*, *ShallowConvNet*, *MBEEG_SENet*, *EEGTCNet*) and dataset (*BCI-IV-2a*, *BCI-IV-2b*, *HGD*), we report subject-wise averages of *classification accuracy* and *Cohen's κ* for two conditions: *None* (no preprocessing) and *RDWT*.

Our goal is to isolate the contribution of RDWT independently of architectural specifics: by comparing each model to its own baseline on the same participants, we directly assess whether RDWT systematically improves accuracy. Beyond aggregate statistics, we also examine subject-level variability to identify regimes where RDWT is most effective (e.g., recordings with pronounced non-stationarity or localized noise) and where gains are naturally limited (e.g., near-ceiling performance). The following subsections detail these results per dataset, highlighting consistent patterns of improvement and noting cases in which RDWT yields negligible or mixed changes, thereby providing an evidence-based view of when this preprocessing is most beneficial.

A. EFFECT OF RDWT PREPROCESSING ON MODEL PERFORMANCE (BCI-IV-2A)

Table 1 reports a paired comparison (with/without RDWT) across four baseline architectures on *BCI-IV-2a*. On average, RDWT yields *consistent* improvements for all models considered, although the magnitude is clearly architecture-dependent. *EEGTCNet* exhibits the largest gain in average accuracy (+4.44 pp; 64.35% $\rightarrow$ 68.79%) together with the largest increase in Cohen's κ (+0.059). *MBEEG_SENet* follows with a +2.23 pp gain (70.49% $\rightarrow$ 72.72%; κ +0.030), while *EEGNet* improves by +2.08 pp (68.02% $\rightarrow$ 70.10%; κ +0.027). *ShallowConvNet* shows a smaller but positive shift (+0.58 pp; 65.74% $\rightarrow$ 66.32%; κ +0.008).

A subject-level inspection reveals heterogeneous effects that clarify where RDWT is most beneficial. For *EEGTCNet*, gains are pronounced on S01 (+12.16 pp) and S04 (+10.07 pp), with additional improvements on S03 (+4.51 pp), S05 (+4.86 pp), and S06 (+3.47 pp), and only a marginal decrease on S07 (−0.34 pp). *MBEEG_SENet* shows a marked jump on S05 (+22.22 pp) and smaller increases on S09 (+2.43 pp), with modest declines on S03/S04 (−1.04/−1.38 pp) and S07 (−3.13 pp). *EEGNet* benefits notably on S09 (+10.07 pp), S06 (+3.82 pp), and S04 (+3.47 pp), while dropping on S05 (−2.78 pp) and S08 (−2.08 pp). *ShallowConvNet* exhibits a large improvement on S04 (+12.50 pp), but decreases on S05/S06/S07 (−6.25/−2.43/−4.16 pp), yielding a modest net gain overall.

Taken together, these results indicate that RDWT is most impactful when the backbone can exploit enhanced time–frequency structure - particularly in temporally

TABLE 1. Comparison of different methods with and without the RDWT module on BCI-IV-2a.

Model	Preproc.	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg. $\pm$ SD	Kappa	p
EEGNet	None	77.08	51.74	91.67	54.17	63.89	45.83	83.68	77.08	67.01	68.02 $\pm$ 15.56	0.5740	–
	RDWT	79.86	51.74	92.71	57.64	61.11	49.65	86.11	75.00	77.08	70.10 $\pm$ 15.52	0.6010	0.0534
ShallowConvNet	None	72.92	46.88	83.68	47.92	63.19	40.97	81.94	78.82	75.35	65.74 $\pm$ 16.56	0.5430	–
	RDWT	74.31	48.26	83.33	60.42	56.94	38.54	77.78	79.86	77.43	66.32 $\pm$ 15.86	0.5510	0.5000
MBEEG_SENet	None	82.29	53.82	92.36	65.62	44.79	57.64	86.46	79.51	71.88	70.49 $\pm$ 16.14	0.6060	–
	RDWT	82.64	54.17	91.32	64.24	67.01	57.99	83.33	79.51	74.31	72.72 $\pm$ 12.61	0.6360	0.3896
EEGTCNet	None	63.19	49.65	81.25	50.69	62.50	46.18	80.90	68.75	76.04	64.35 $\pm$ 13.48	0.5250	–
	RDWT	75.35	52.78	85.76	60.76	67.36	49.65	80.56	70.14	76.74	68.79 $\pm$ 12.37	0.5840	0.0039

TABLE 2. Comparison of different methods with DB4 hard and soft module on BCI-IV-2a. None of the wavelet-based preprocessing improves the results.

Model	Config	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg.	Kappa
EEGNET	None	77.08	51.74	91.67	54.17	63.89	45.83	83.68	77.08	67.01	68.02	0.574
	DB4_Soft	61.11	47.57	60.07	52.43	66.32	50.35	71.88	58.33	50.35	57.60	0.435
	DB4_Hard	63.54	44.79	60.42	55.56	62.85	51.74	65.28	63.19	48.61	57.33	0.431
ShallowConvNet	None	72.92	46.88	83.68	47.92	63.19	40.97	81.94	78.82	75.35	65.74	0.543
	DB4_Soft	36.81	23.96	34.03	28.12	51.04	28.47	52.43	48.96	51.04	39.43	0.192
	DB4_Hard	37.50	24.65	35.42	26.39	53.82	30.90	63.54	57.64	54.51	42.71	0.236
MBEEG_SENet	None	82.29	53.82	92.36	65.62	44.79	57.64	86.46	79.51	71.88	70.49	0.606
	DB4_Soft	62.15	42.01	59.72	57.99	67.36	51.04	66.32	62.15	53.12	57.99	0.440
	DB4_Hard	63.19	40.62	61.81	58.68	64.93	48.61	71.88	65.97	55.90	59.07	0.454
EEGTCNet	None	63.19	49.65	81.25	50.69	62.50	46.18	80.90	68.75	76.04	64.35	0.525
	DB4_Soft	52.08	34.38	56.94	44.10	62.85	48.61	65.97	55.90	57.64	53.16	0.376
	DB4_Hard	51.39	40.28	63.89	53.82	64.58	49.65	65.62	56.94	51.04	55.25	0.403

expressive models such as TCN-style stacks (EEGTC-Net) - and for subjects whose recordings present greater localized noise. Conversely, compact CNNs with limited temporal modeling capacity tend to realize smaller average benefits, with mixed subject-wise responses. In practical terms, RDWT emerges as a low-overhead, architecture-aware preprocessing step that can yield meaningful accuracy and agreement improvements on BCI-IV-2a, especially for models and subjects that can capitalize on its multi-resolution representation.

RDWT demonstrated statistically significant improvement on the EEGTCNet model, where the mean accuracy increase was highly reliable ($p = 0.0039$). The EEGNet model showed a substantial positive trend, achieving the highest p -value near the common $\alpha = 0.05$ threshold ($p = 0.0534$), indicating a strong statistical benefit. Conversely, the gains observed for ShallowConvNet ($p = 0.5000$) and MBEEG_SENet ($p = 0.3896$) were not statistically significant. This pattern of results underscores the architecture-aware nature of RDWT, confirming that its feature enhancement capabilities are maximized when coupled with models that possess strong temporal processing capabilities (like EEGTCNet).

The use of DB4 Soft and Hard as an alternative wavelet-based preprocessing, in the Table 2, reveals that

does not improve performance: for all backbones, the subject-averaged accuracy and Cohen's κ are lower, or at best comparable, to the no-preprocessing strategy (i.e., None). This indicates that non-adaptive wavelet filtering may suppress subject-specific discriminative components, yielding limited benefit for MI-EEG decoding.

B. EFFECT OF RDWT PREPROCESSING ON MODEL PERFORMANCE (BCI-IV-2B)

Table 3 indicates that the impact of RDWT on BCI-IV-2b is modest due to the effect of ceiling performance on several subjects ($\geq 99\%$). In average accuracy, RDWT yields small but consistent gains for EEGNet (+0.21 pp; 95.85% $\rightarrow$ 96.06%) and EEGTCNet (+0.28 pp; 95.81% $\rightarrow$ 96.09%), and a minor improvement for ShallowConvNet (+0.09 pp; 95.85% $\rightarrow$ 95.94%). MBEEG_SENet shows a slight increase in average accuracy (+0.14 pp; 96.39% $\rightarrow$ 96.53%) but a decrease in agreement (κ : 0.671 $\rightarrow$ 0.646, -0.025). Agreement gains are otherwise positive for EEGNet (κ +0.044), EEGTCNet (κ +0.077), and ShallowConvNet (κ +0.005).

Subject-wise patterns reveal where RDWT is most beneficial. For EEGNet, S09 exhibits a marked improvement (+5.62 pp; 93.51% $\rightarrow$ 99.13%), with S02 and S06

TABLE 3. Comparison of different methods with and without the RDWT module on BCI-IV-2b.

Model	Preproc.	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg. $\pm$ SD	Kappa	p
EEGNet	None	98.58	99.10	98.19	84.44	92.95	99.53	97.41	98.91	93.51	95.85 $\pm$ 4.91	0.5960	–
	RDWT	99.29	100.00	98.55	80.00	92.31	100.00	97.04	98.19	99.13	96.06 $\pm$ 6.47	0.6400	0.4102
ShallowConvNet	None	99.65	99.10	97.83	82.22	91.67	99.06	95.19	99.28	98.70	95.85 $\pm$ 5.73	0.6180	–
	RDWT	99.65	99.10	96.74	86.67	91.67	99.06	93.33	98.55	98.70	95.94 $\pm$ 4.47	0.6230	0.6425
MBEEG_SENet	None	98.94	98.20	99.28	82.22	92.95	100.00	97.04	98.91	100.00	96.39 $\pm$ 5.74	0.6710	–
	RDWT	98.94	99.10	97.83	84.44	92.95	100.00	97.04	98.91	99.57	96.53 $\pm$ 5.00	0.6460	0.3575
EEGTCNet	None	98.23	99.10	97.10	84.44	90.38	100.00	97.41	98.19	97.40	95.81 $\pm$ 5.07	0.5890	–
	RDWT	99.65	98.20	98.19	82.22	91.67	99.53	97.04	99.64	98.70	96.09 $\pm$ 5.76	0.6660	0.2129

reaching 100%, while S04 decreases (–4.44 pp). EEGTCNet gains on S01 (+1.42 pp), S03 (+1.09 pp), S05 (+1.29 pp), S08 (+1.45 pp), and S09 (+1.30 pp), but drops on S04 (–2.22 pp) and marginally on S06/S07. ShallowConvNet benefits chiefly on S04 (+4.45 pp) amid small declines on S03 (–1.09 pp), S07 (–1.86 pp), and S08 (–0.73 pp). MBEEG_SENet shows mixed changes, improvements on S02 (+0.90 pp) and S04 (+2.22 pp) contrasted with decreases on S03 (–1.45 pp) and S09 (–0.43 pp), which plausibly explains the slight rise in accuracy alongside a lower κ .

Overall, RDWT can deliver small accuracy and agreement gains for backbones able to exploit additional time–frequency structure (notably TCN-based EEGTCNet and, to a lesser extent, EEGNet). In near-ceiling regimes typical of BCI-IV-2b, its average benefits are limited and architecture-specific, with occasional subject-level improvements offset by minor degradations on others.

C. EFFECT OF RDWT PREPROCESSING ON MODEL PERFORMANCE (HGD DATASET)

Table 4 shows that RDWT produces consistent average gains also on HGD, with clear architecture-dependent effects. In terms of average accuracy, the largest improvement is observed for MBEEG_SENet (+1.65 pp; 88.61% $\rightarrow$ 90.26%), accompanied by a higher agreement (κ : 0.848 $\rightarrow$ 0.870, +0.022). EEGNet and EEGTCNet also benefit, with accuracy shifts of +0.76 pp (87.32% $\rightarrow$ 88.08%; κ +0.010) and +0.54 pp (86.60% $\rightarrow$ 87.14%; κ +0.008), respectively. ShallowConvNet exhibits a near-null change (+0.22 pp; 87.05% $\rightarrow$ 87.27%; κ +0.003). Subject-level analyses reveal that RDWT can be markedly beneficial on difficult cases. The most prominent example is S14, where accuracy increases for EEGNet (+21.25 pp; 57.50% $\rightarrow$ 78.75%), MBEEG_SENet (+21.26 pp; 60.62% $\rightarrow$ 81.88%), and ShallowConvNet (+6.26 pp). On S11, EEGTCNet improves by +7.50 pp, whereas MBEEG_SENet remains unchanged. On S6, RDWT benefits EEGTCNet (+3.13 pp) and MBEEG_SENet (+1.26 pp). These results suggest that RDWT is most effective when additional time–frequency structure aids temporal modeling; conversely, when recordings are already clean or the backbone captures rich multi-scale features, net improvements are limited. In absolute terms, the best overall

configuration on HGD is MBEEG_SENet with RDWT (90.26% average accuracy; κ = 0.870).

Table 5 reports wall-clock training time (in minutes) for each backbone across datasets with and without the RDWT front-end. On BCI-IV-2a and BCI-IV-2b, RDWT does not introduce a material overhead, the absolute differences are typically $\leq$ 0.3 minutes per run. On HGD, changes remain small for most models (within $\pm$ 1.6 minutes), with the exception of MBEEG_SENet, which shows a marked reduction (–12.8 minutes). Overall, the computational footprint of RDWT is comparable to the baseline.

For each dataset–model pair and for each subject, we release a Jupyter notebook that enables transparent inspection of the training and test behaviour. Specifically, every notebook includes: (i) *training* and *validation* accuracy and loss traces across epochs; and (ii) *test*-phase diagnostics comprising a per-class bar plot and the confusion matrix. These artifacts allow readers to replicate runs and examine subject-level dynamics beyond the aggregate tables.

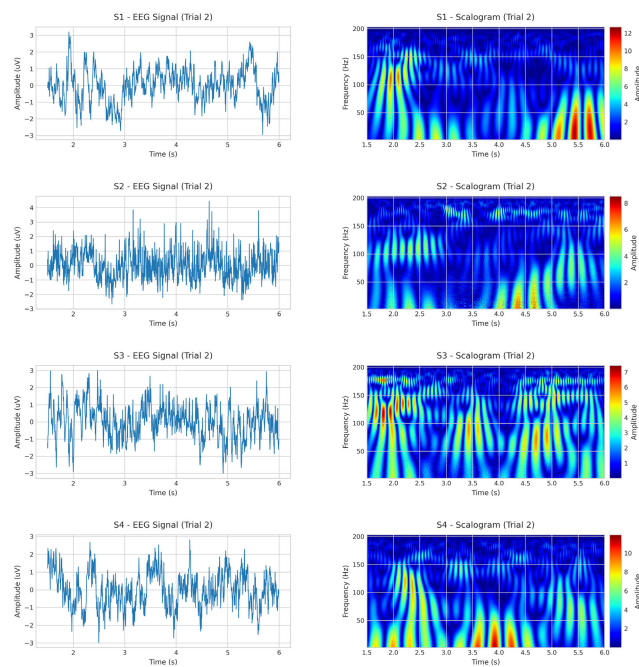
D. SCALOGRAM-BASED ANALYSIS OF EEG SIGNALS

Figure 1 presents a comparative visualization of EEG signals and their corresponding scalograms for the “Left Hand Movement” class, recorded from the Cz channel during the second trial across four subjects (S1–S4). The left panels show raw EEG signals in the time domain, while the right panels depict their time-frequency representations obtained using the continuous wavelet transform (CWT). While the EEG amplitudes are similar in range, the scalograms reveal subject-specific spectral patterns, particularly in the 10–30 Hz and 100–150 Hz bands. These differences may reflect inter-subject variability in motor intention processing and support the use of time-frequency features for improving classification performance in motor imagery tasks.

Time–frequency representations offer an intuitive tool for visual inspection of EEG signals in motor imagery tasks. However, a detailed scalogram-based analysis revealed that frequency-domain patterns alone do not provide sufficient discriminative information across classes or subjects. As illustrated in Figure 2, for three representative subjects and across three different EEG channels (Cz, C3, and C4), the scalograms display highly similar spectral structures,

TABLE 4. Comparison of different methods with and without the RDWT module on HGD (14 subjects). We report subject-wise accuracies (S1–S14) and inter-subject variability as Avg. $\pm$ SD.

Model	Preproc.	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Avg. $\pm$ SD	Kappa	<i>p</i>
EEGNet	None	89.38	81.25	97.50	95.62	90.00	94.38	91.19	87.50	95.62	86.25	76.88	93.12	86.25	57.50	87.32 $\pm$ 10.34	0.8310	–
	RDWT	87.50	83.12	93.75	93.12	90.62	88.12	90.57	90.00	95.62	88.12	71.88	91.88	90.00	78.75	88.08 $\pm$ 6.34	0.8410	0.6368
ShallowConvNet	None	88.75	87.50	96.25	95.62	88.75	91.88	88.05	85.62	95.62	85.62	68.75	91.88	76.25	78.12	87.05 $\pm$ 7.94	0.8270	–
	RDWT	86.88	90.00	98.75	96.88	90.62	90.62	86.79	80.00	96.25	89.38	61.25	93.75	76.25	84.38	87.27 $\pm$ 9.77	0.8300	0.2528
MBEEG_SENet	None	92.50	84.38	97.50	96.25	91.25	90.62	91.82	89.38	95.62	90.62	76.88	93.75	89.38	60.62	88.61 $\pm$ 9.58	0.8480	–
	RDWT	91.88	91.88	95.62	94.38	94.38	91.88	91.19	92.50	95.00	86.25	76.88	91.88	88.12	81.88	90.26 $\pm$ 5.35	0.8700	0.4169
EEGTCNet	None	87.50	90.62	95.62	90.62	90.00	85.62	93.08	87.50	95.00	85.00	65.62	93.12	84.38	68.75	86.60 $\pm$ 8.98	0.8210	–
	RDWT	86.25	90.62	94.38	92.50	90.00	88.75	88.68	88.12	95.00	88.12	73.12	90.62	90.62	63.12	87.14 $\pm$ 8.63	0.8290	0.2969

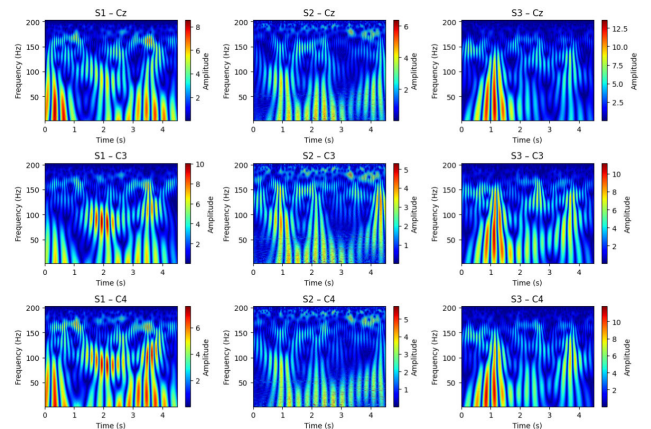
**FIGURE 1.** Comparison of EEG signals and scalograms (Left Hand Movement, Cz, Trial 2) for subjects S1–S4. Left: time-domain signals; right: time-frequency maps. The spectral differences reflect inter-subject variability in motor-related activity.

regardless of the channel considered. This suggests a high level of redundancy in the frequency response across channels, which justifies our decision to retain a single representative channel in the subsequent analysis.

Furthermore, Figure 3 compares the spectral responses of the first four subjects of the BCI Competition IV-2a dataset while imagining the four movements (same channel). The Figure is a visualization of the first trial for each motor imagery class (left hand, right hand, foot, tongue) across four subjects (S3, S7, S2 and S6) from the BCI-IV-2a dataset. The first two (i.e., S3 and S7) are the subjects providing the highest classification accuracies while the last two (i.e., S2 and S6) are the ones with the worst classification accuracies. Consistently with the previous observation, the spectrograms do not exhibit distinctive or

TABLE 5. Wall-clock training time per model across datasets (minutes).

Model	Dataset	None	RDWT
EEGNet	BCI-IV-2a	9.0	8.7
	BCI-IV-2b	8.1	7.9
	HGD	33.8	35.2
ShallowConvNet	BCI-IV-2a	8.7	8.8
	BCI-IV-2b	7.5	7.2
	HGD	31.9	30.3
MBEEG_SENet	BCI-IV-2a	12.8	12.7
	BCI-IV-2b	12.4	12.2
	HGD	56.2	43.4
EEGTCNet	BCI-IV-2a	9.9	9.9
	BCI-IV-2b	9.6	9.4
	HGD	34.7	35.6

**FIGURE 2.** Scalograms for S1–S3 across channels Cz, C3, C4 during left-hand motor imagery. The high spectral similarity across channels suggests redundancy, supporting the choice of a single representative channel for further analysis.

consistent patterns, either within or across subjects and classes. This visual evidence aligns with the relatively low classification performance obtained using frequency-domain features, supporting the conclusion that purely spectral information is insufficient for reliable discrimination in this context.

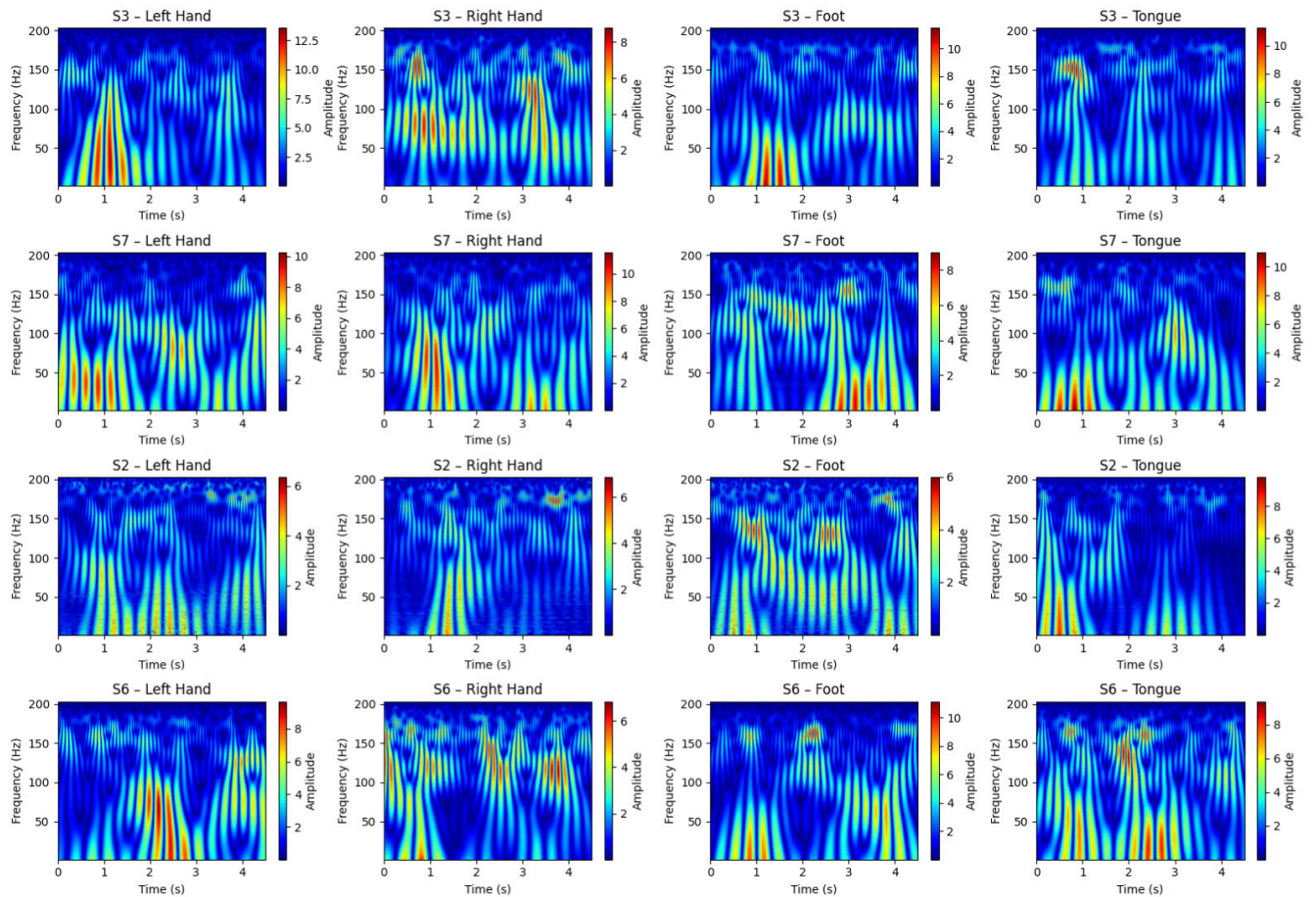


FIGURE 3. Scalogram visualization of the first trial for each motor imagery class (left hand, right hand, foot, tongue) in four BCI-IV-2a subjects: S3 and S7 (best classification) vs. S2 and S6 (worst). Rows: subjects; columns: classes. The absence of consistent patterns highlights the limited discriminative power of frequency-domain features.

Our observations, corroborated by recent studies [7], [26], suggest that frequency-based analyses may fail to capture the dynamic temporal variations inherent in EEG data, leading to suboptimal classification performance. For example, in [26], the authors employed CNNs to investigate the impact of generating topological representations from motor imagery EEG signals in both the time and frequency domains. Their findings indicate that maps derived from the time domain significantly outperform those constructed using frequency-domain information, highlighting the superiority of time-resolved features for capturing relevant neural dynamics in motor imagery tasks. Finally, As reported in [7], studies utilizing raw EEG signal inputs on the most frequently used motor imagery dataset (BCI Competition IV-2a) achieved an average accuracy of 77.65%. This performance slightly surpasses that of approaches based on extracted features (77.02%), spectral images (76.80%), and topological maps (76.5%). Similarly, for the second most commonly employed dataset (BCI Competition IV-2b), models trained directly on raw EEG signals attained an average accuracy of 83.22%, outperforming those relying on extracted features (81.56%) and spectral image inputs (82.07%).

Consequently, these results advocate for the adoption of time-resolved and adaptive signal processing methods that better accommodate the non-stationary nature of EEG signals in brain-computer interface applications.

Figure 4 illustrates the inter-subject correlation matrices for four distinct motor imagery tasks, left hand, right hand, feet, and tongue, based on the maximum normalized cross-correlation values computed from EEG signals at channel Cz during Trial 2. Also in this case the correlation matrices are related to the two top-classification-accuracies subjects (i.e., S3 and S7) and to the two worst-classification-accuracies subjects (i.e., S2 and S6). A key observation from these heatmaps is the consistently high self-correlation (1.00) for each subject across all tasks, as expected. More critically, the figure highlights the varying degrees of cross-subject correlation. Notably, the correlations between subjects S2 and S6 are consistently lower compared to the correlations involving S3 and S7, regardless of the motor imagery task. For instance, in the “Left Hand” task, the cross-correlation between S2 and S6 is 0.13, significantly lower than the 0.20 correlation between S3 and S7 for the same task. This pervasive pattern of lower cross-subject correlations for

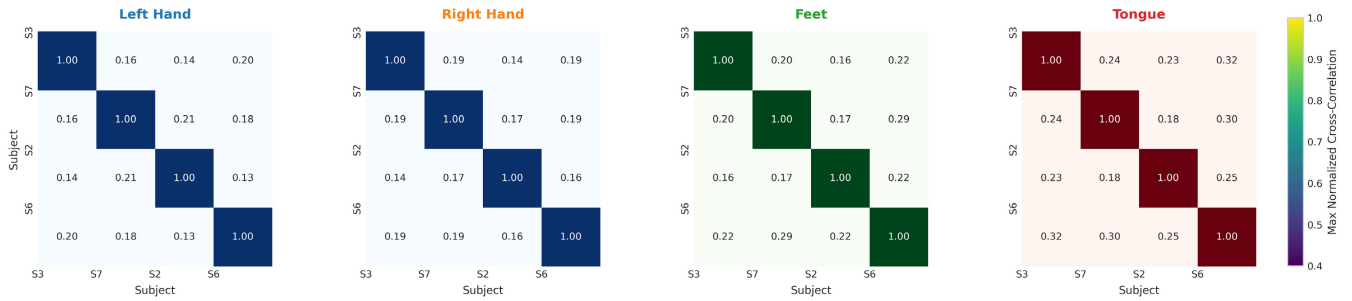


FIGURE 4. Inter-subject correlation matrices (Trial 3, Cz) for four motor imagery tasks. Matrices show pairwise maximum normalized cross-correlations between S3, S7, S2, and S6. Higher off-diagonal values indicate greater cross-subject similarity.

S2 and S6 across all tasks suggests a reduced consistency in their brain activity patterns compared to S3 and S7. This finding is particularly pertinent when considering the previously discussed challenges in classifying data from S2 and S6 (as highlighted in the context of Figure 6), as it directly indicates a lower degree of shared neural features that a classification algorithm could exploit across these more challenging subjects. The overall implication is that the lack of robust inter-subject commonalities, especially for subjects like S2 and S6, poses a significant hurdle for achieving high classification accuracies in brain-computer interface systems that rely on generalized models or feature extraction techniques.

VI. DISCUSSION AND CONCLUSION

This study assessed the impact of RDWT preprocessing on EEG decoding in a paired (with/without) setting across four widely used backbones, EEGNet, ShallowConvNet, MBEEG_SENet, and EEGTCNet, on three benchmark motor-imagery datasets (BCI-IV-2a, BCI-IV-2b, HGD). Our analysis isolates the contribution of RDWT as a plug-in, architecture-agnostic step placed upstream of otherwise unchanged classifiers. Briefly, the answers to our RQs are:

- **A1 (to RQ1).** RDWT yields positive, architecture-dependent gains. On BCI-IV-2a, the average accuracy increases are +4.44 pp for EEG-TCNet (with $\kappa+0.059$), +2.23 pp for MBEEG_SENet ($\kappa+0.030$), +2.08 pp for EEGNet ($\kappa+0.027$), and +0.58 pp for ShallowConvNet ($\kappa+0.008$). On BCI-IV-2b, improvements are necessarily modest due to near-ceiling performance but remain positive for EEGNet and EEG-TCNet. On HGD, effects are modest-to-positive overall, largest for MBEEG_SENet (+1.65 pp; $\kappa+0.022$).
- **A2 (to RQ2).** RDWT helps most when the backbone can exploit enhanced time–frequency structure (e.g., temporal/TCN-style stacks) and for difficult or non-stationary signals; benefits are limited in near-ceiling regimes or with compact CNNs that already capture multi-scale structure.

A. SUMMARY OF FINDINGS

RDWT's effect is unequivocally *architecture- and dataset-dependent*. On BCI-IV-2a (Table 1), RDWT yields clear

average gains for EEGTCNet (+4.44 pp; $\kappa+0.059$) and MBEEG_SENet (+2.23 pp; $\kappa+0.030$), with smaller but positive shifts for EEGNet (+2.08 pp; $\kappa+0.027$) and ShallowConvNet (+0.58 pp; $\kappa+0.008$). On BCI-IV-2b (Table 3), where several subjects operate near ceiling, improvements are necessarily modest: EEGTCNet (+0.28 pp; $\kappa+0.077$) and EEGNet (+0.21 pp; $\kappa+0.044$) benefit slightly; ShallowConvNet shows a near-null change (+0.09 pp; $\kappa+0.005$); MBEEG_SENet exhibits a small accuracy increase (+0.14 pp) alongside a lower agreement (-0.025κ). On HGD (Table 4), average effects are modest-to-positive across all four models, with MBEEG_SENet showing the largest gain (+1.65 pp; $\kappa+0.022$), followed by EEGNet (+0.76 pp; $\kappa+0.010$), EEGTCNet (+0.54 pp; $\kappa+0.008$), and ShallowConvNet (+0.22 pp; $\kappa+0.003$).

B. WHEN AND WHY RDWT HELPS

Subject-wise patterns reveal that RDWT can be decisively beneficial on difficult cases (e.g., S14 in HGD, with double-digit gains for multiple backbones), suggesting that the added time–frequency structure mitigates noise and improves rhythm isolation. Architectures with stronger temporal modeling capacity (e.g., TCN-style stacks such as EEGTCNet) tend to leverage RDWT more consistently, while compact CNNs display smaller, heterogeneous responses that depend on subject and dataset headroom.

C. PRACTICAL GUIDANCE

RDWT is a low-overhead, plug-in preprocessing suitable for pipelines that (i) face pronounced inter/intra-subject variability, (ii) rely on temporal modeling to capture task-relevant dynamics, and/or (iii) are not already operating at ceiling. In near-ceiling regimes (typical of BCI-IV-2b), expected gains are small; selective or subject-adaptive activation of RDWT may be preferable to blanket application.

D. LIMITATIONS AND FUTURE WORK

Average improvements, while consistent in direction for several model–dataset pairs, are modest; future studies should increase cohort sizes and report paired statistical testing. Automating RDWT hyperparameters (rational dilation factors, mother wavelet choice) via data-driven or

differentiable selection, as well as learning subject- or session-adaptive policies that trigger RDWT only when predicted to help, are promising directions. Extending the analysis beyond motor imagery will further clarify the scope of RDWT in broader EEG paradigms.

E. ADDITIONAL FUTURE DIRECTIONS

Building on the above, we will transition from a static front-end to a **subject-adaptive RDWT** that is enabled only when beneficial, learning a binary gate and selecting both the number of levels and the rational dilations from validation-time gain estimates or lightweight meta-learning policies; this will make explicit the accuracy/ κ -latency trade-offs. We will also address **online/real-time BCI** by profiling preprocessing time (t_{RDWT}), backbone time (t_{net}), and end-to-end latency ($t_{tot} = t_{RDWT} + t_{net}$) under streaming constraints on CPU/edge/GPU hardware, and by studying buffering and RDWT-inference overlap to guarantee predictable closed-loop control. Finally, to strengthen **robustness and domain shift** tolerance, we will integrate RDWT with alignment and transfer techniques to mitigate cross-session and cross-subject variability, and evaluate resistance to artifacts and out-of-distribution inputs through systematic stress tests.

F. CONCLUSION

RDWT preprocessing is not a universal remedy, but it is a practical and sometimes impactful addition that can improve robustness and agreement for specific backbones and challenging subjects. As such, it merits consideration as an optional, architecture-aware component in modern EEG deep learning-based decoding pipelines.

AUTHOR CONTRIBUTIONS

Marco Siino led the conceptualization of the study, designed the methodology, developed the software, and conducted the investigation. He also contributed to the original draft writing, revision, supervision, and project administration. Giuseppe Bonomo contributed to software development, data curation, validation, investigation, and writing of the original draft, including result visualization. Rosario Sorbello assisted with validation, supervision, and drafting the manuscript. Ilenia Tinnirello provided resources, supervised the project, contributed to methodology discussions, manuscript review, and secured funding.

REPRODUCIBILITY STATEMENT

The code for this article is available at github.com/marco-siino/bci-gpt and github.com/Bonomo31/The-Impact-of-the-RDWT-on-EEG-Models

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